



CVR COLLEGE OF ENGINEERING

RISC-V Custom Instruction Multiply-Accumulate Hardware

Work done by

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Introduction

- RISC-V ISA is an open and extensible architecture that supports custom instructions for application-specific performance enhancement.
- Autonomous swarm drones require frequent and real-time multiply-accumulate (MAC) computations for collision avoidance and coordination.
- Conventional RV32IM processors execute MAC using separate multiplication and addition instructions, increasing latency, instruction count, and energy consumption.
- To address this, a custom MAC instruction is proposed with a dedicated hardware MAC unit integrated into the execute stage.
- The MAC unit uses a Booth multiplier and a dedicated accumulator adder to perform $rd = rd + (rs1$



Objectives

- Design and implement a 16×16 -bit MAC Unit using a Radix-4 Modified Booth Multiplier and Accumulate Adder in the Execution stage for single-cycle multiply-accumulate operation.
- Extend the RV32IM instruction set with a custom R-type MAC instruction using RISC-V custom-0 opcode space without breaking backward ISA compatibility.
- Integrate the MAC unit into a complete 5-stage pipelined RV32IM processor (IF $\rightarrow$ ID $\rightarrow$ EX $\rightarrow$ MEM $\rightarrow$ WB) alongside the standard ALU and M-extension Multiplier/Divider Unit.
- Implement pipeline hazard handling using a Hazard Detection Unit for load-use stalls and a Forwarding Unit to resolve data hazards including MAC accumulator forwarding.
- Validate the complete RTL design in Verilog HDL through simulation covering MAC instruction, forwarding chains, branch resolution, and load/store operations.



Hardware Components Required

- An FPGA (Field-Programmable Gate Array) Kit is a hardware development board used to design, prototype, and test digital circuits and embedded systems. These kits allow users to program the FPGA chip to implement custom logic circuits, processors, and embedded applications.
- Power supply: Provides Power to the FPGA board

FPGA BOARD

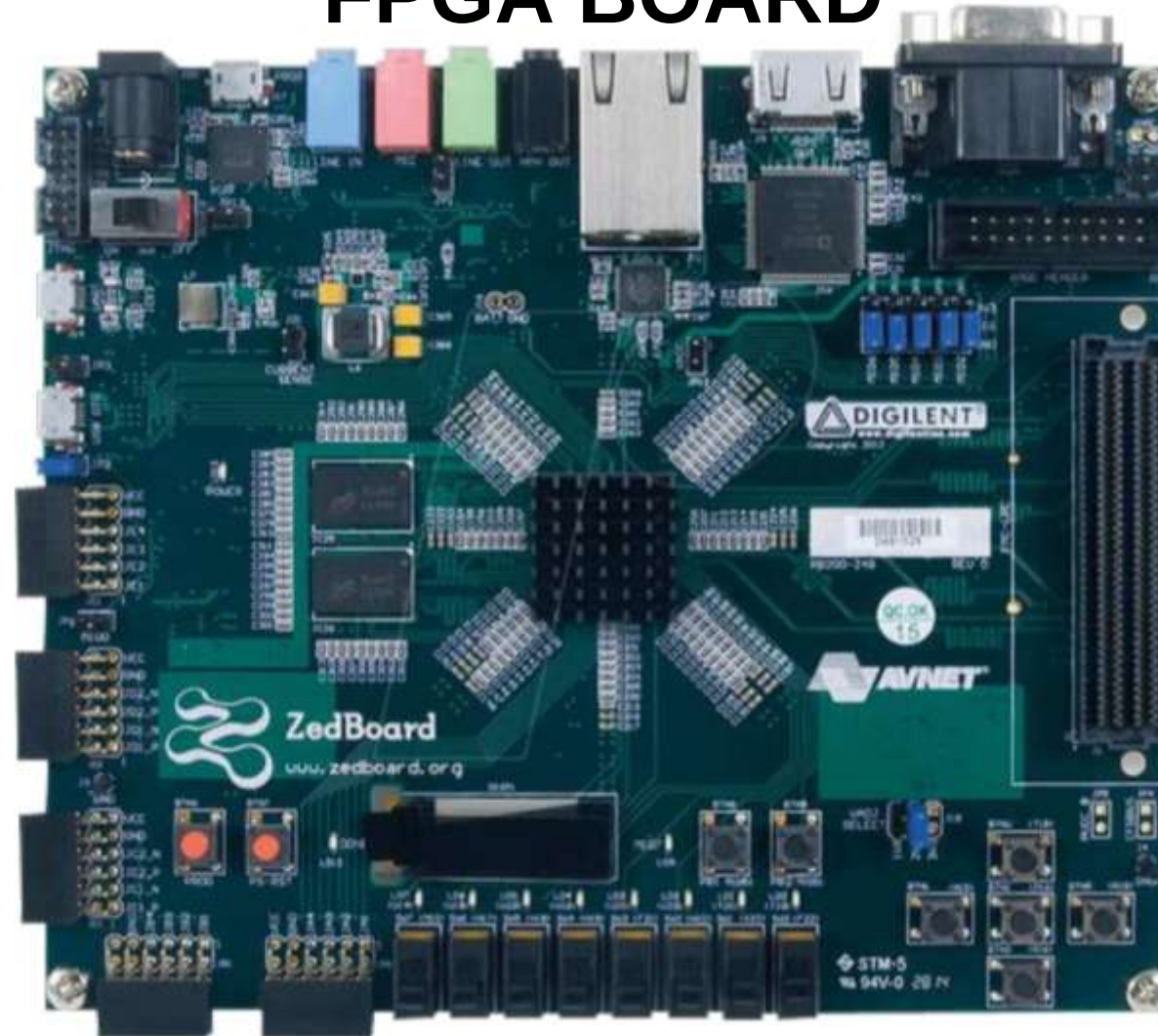
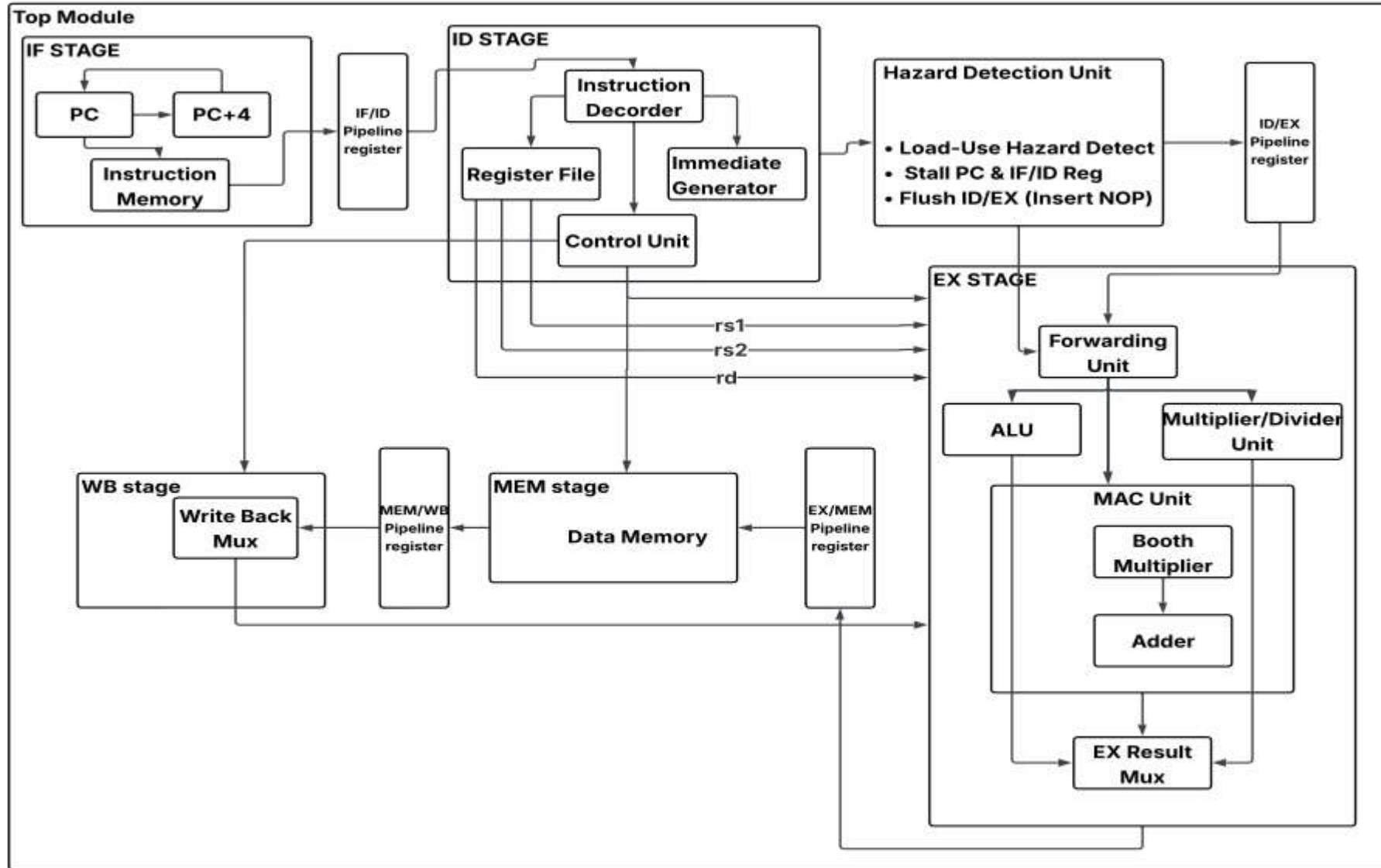


Fig.31 zynq-7000 FPGA Board

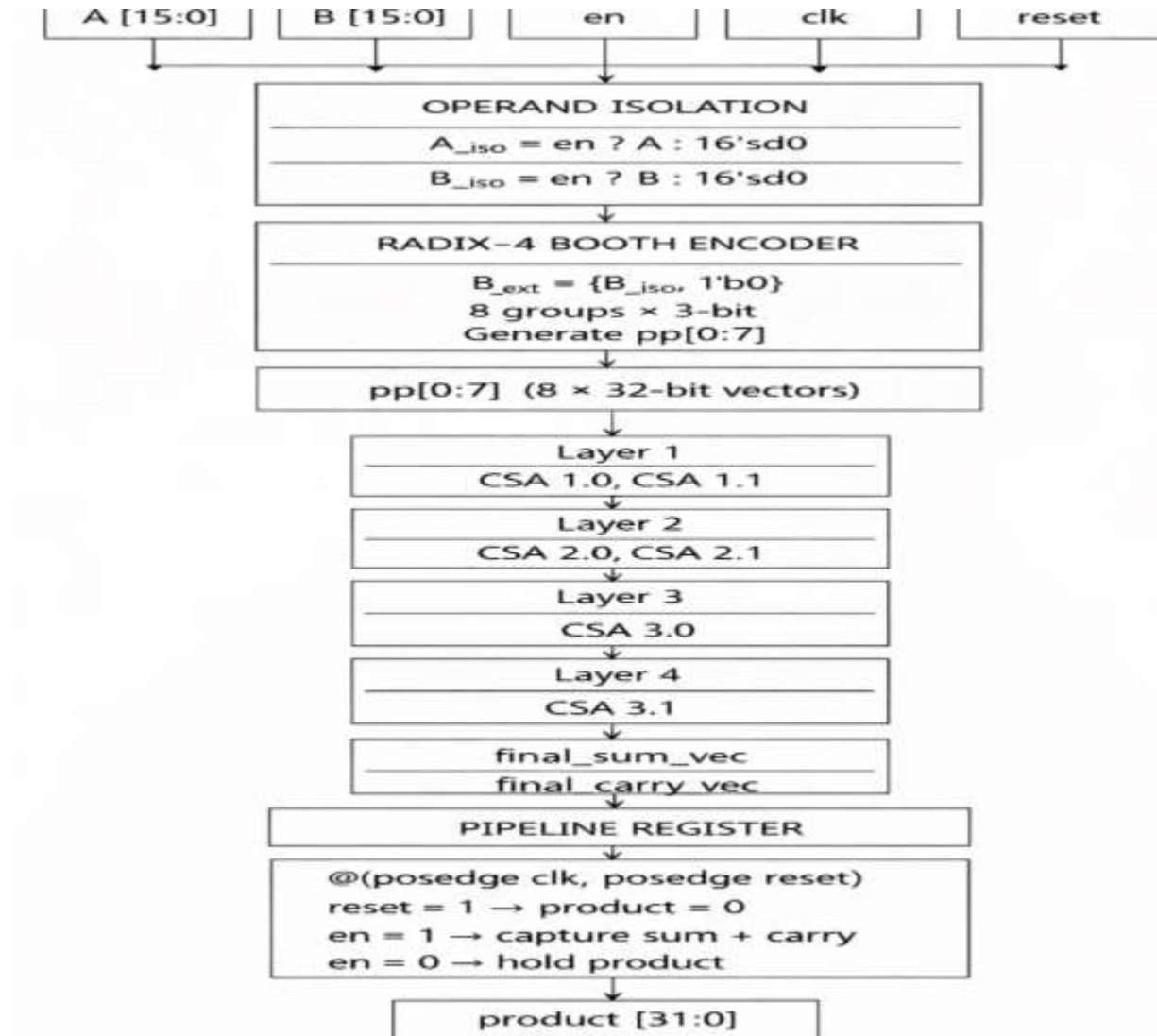
Software Requirement

- Xilinx Vivado is a software tool used to design digital circuits, especially for FPGA (Field-Programmable Gate Array) and SoC (System on Chip) development
- Verilog HDL is used to design the RISC-V core, custom MAC unit, and modified execution datapath.
- Simulate the Design : Test your Verilog code to see if the logic works correctly before implementing it on hardware.
- Synthesize the Code: Convert Verilog into a form that an FPGA can understand.
- Implement & Generate Bitstream : This final step prepares the design for loading onto an FPGA chip.
- Debug & Analyze: Vivado has tools like waveform viewers to help you check and fix errors in your design.
- Then we use the Cadence Tool for Synthesis the design code with 90nm Technology to get to know the Area , Power and Timing of the design.
- We use the Genus for Synthesis.

Block Diagram

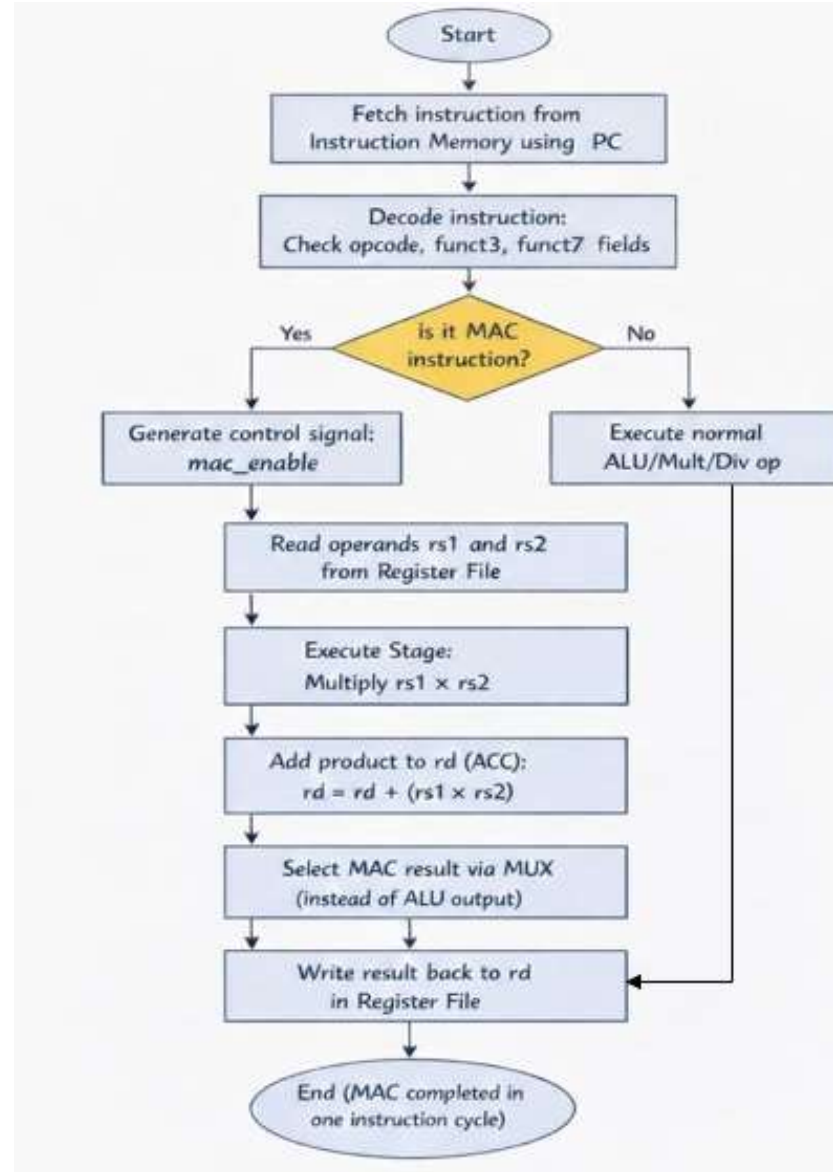


Working of Modified Booth Multiplier



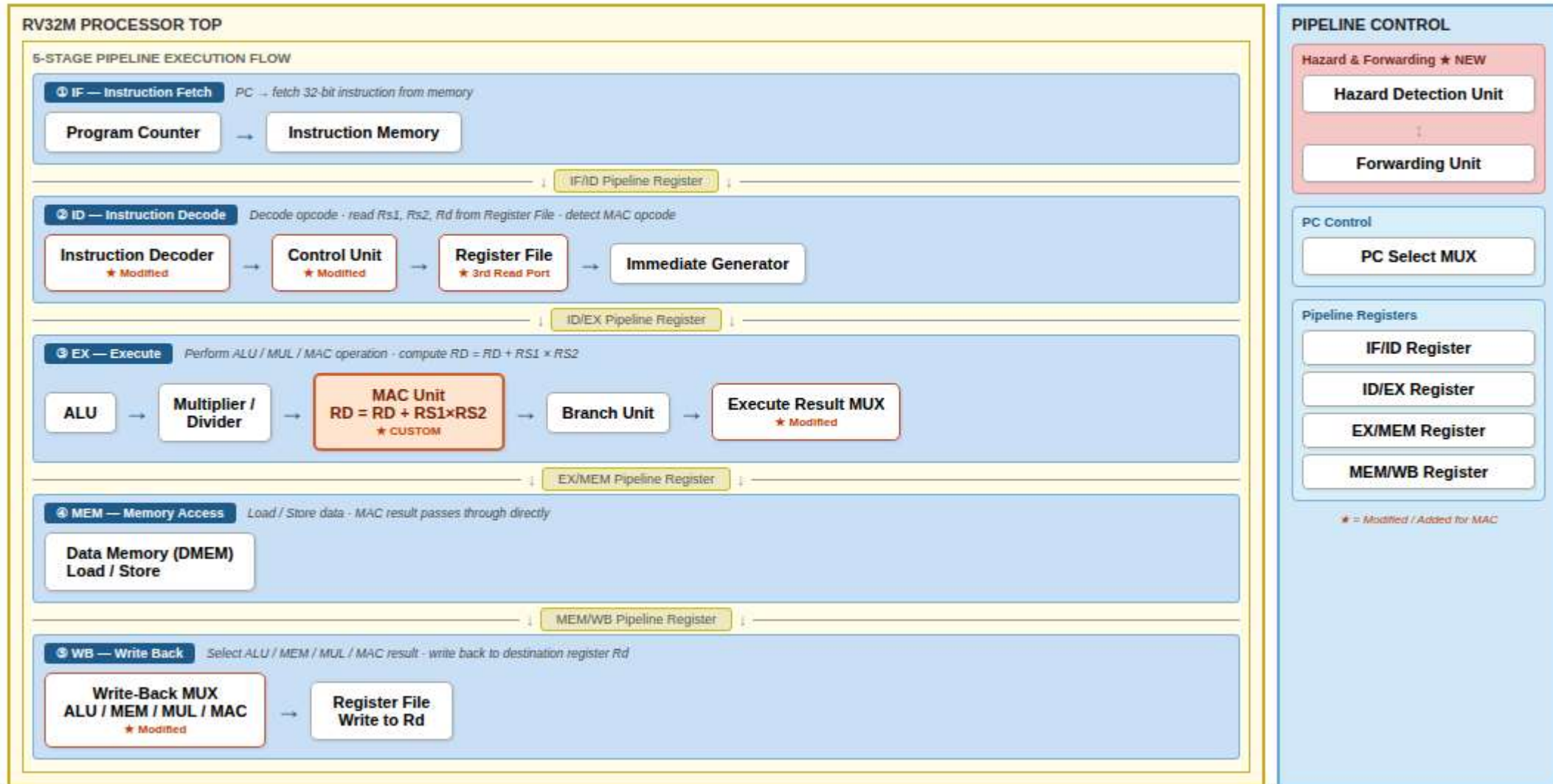
B_ext[i+2:i]	Booth Code	Partial Product
000	+0	0
001	+1	+A << (2*i)
010	+1	+A << (2*i)
011	+2	+A << (2*i + 1)
100	-2	-A << (2*i + 1)
101	-1	-A << (2*i)
110	-1	-A << (2*i)
111	+0	0

Working principle



RISC-V RV32M Processor with Custom MAC Instruction — Execution Flow

TOP MODULE : riscv_mac_top



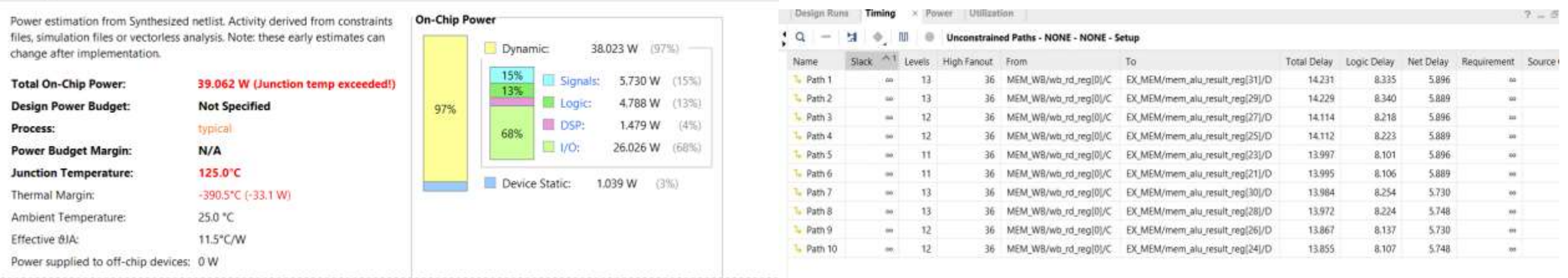
Comparation

Metric	Conventional Multiplication	Normal Booth Multiplier with Conventional Adder	Modified Booth Multiplier with CSA Adders	Trend / Observation
Total On-Chip Power	37.905 W	35.058 W	38.575 W	Power remains high due to I/O switching.
Logic Delay (Max)	6.456 ns	7.340 ns	3.741 ns	42% improvement in logic speed with CSA.
Total Delay (Max)	11.916 ns	13.138 ns	8.735 ns	CSA version is the fastest overall.
Slice LUTs	399	473	443	CSA is more area-efficient than "Normal Booth."
Junction Temp	125.0°C	125.0°C	125.0°C	All exceed thermal limits (Critical).
Critical Path Start	A[1]	B[8]\$	en	High fanout on en is the new bottleneck.

MAC Unit Comparison

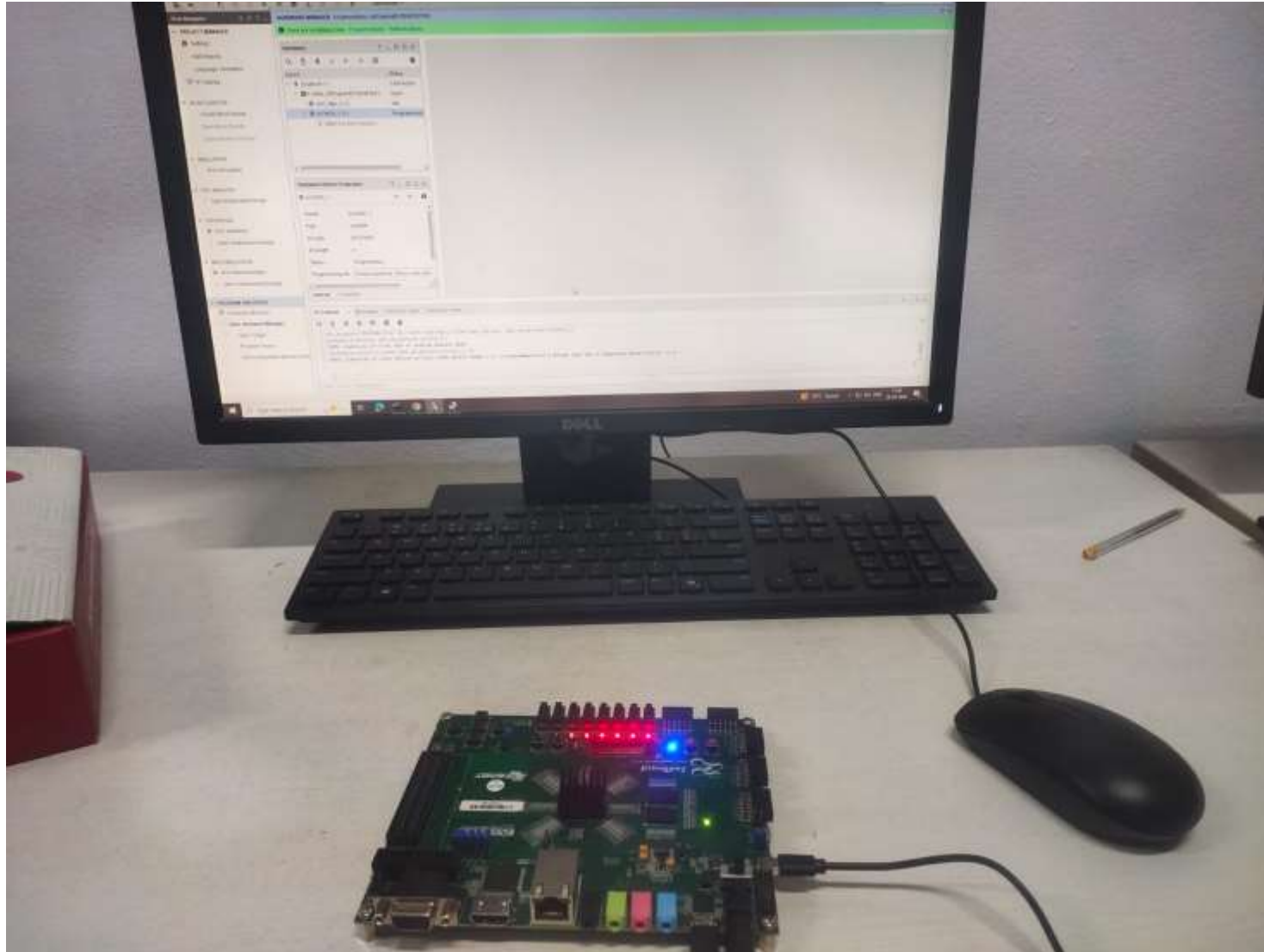
Feature	MAC Pure Combinational	MAC 1-Cycle (Comb. Mult + Seq. Acc)	MAC 2cycle_Fully Sequential
Total On-Chip Power	15.75 W (Lowest)	21.632 W	21.53 W
Dynamic Power	14.711 W (93%)	20.592 W (95%)	20.490 W (95%)
Total Delay (Max)	51.951 ns (Slowest)	11.255 ns	9.511 ns (Fastest)
Logic Levels	77 Levels	16 Levels	14 Levels
Slice LUTs	523	522	532
Slice Registers	0	33	65
Junction Temp	125.0°C (Exceeded)	125.0°C (Exceeded)	125.0°C (Exceeded)
Thermal Margin	-121.7°C	-189.5°C	-188.3°C

Results of RV32IM Processor integrated with MAC unit



Name	Slice LUTs (53200)	Slice Registers (106400)	F7 Muxes (26600)	F8 Muxes (13300)	DSPs (220)	Bonded IOB (200)	BUFGCTRL (32)
✖ riscv_mac_processor	1515	576	165	32	5	104	1
DMEM (data_memory)	128	0	64	32	0	0	0
EX_MEM (ex_mem_register)	465	74	23	0	0	0	0
ID_EX (id_ex_register)	144	93	14	0	0	0	0
IF_ID (if_id_register)	27	18	0	0	0	0	0
IMMGEN (immediate_generator)	3	0	0	0	0	0	0
MAC (mac_unit)	508	67	0	0	0	0	0
adder (accumulator_adder)	65	33	0	0	0	0	0
mult (booth_multiplier)	443	32	0	0	0	0	0
MEM_WB (mem_wb_register)	76	70	0	0	0	0	0
MULTIPLIER (multiplier)	30	0	0	0	5	0	0
PC (program_counter)	11	30	0	0	0	0	0
REGFILE (register_file)	128	224	64	0	0	0	0

Implementation on FPGA





Conclusions

- The proposed work focuses on extending the RISC-V processor by integrating a custom Multiply-Accumulate (MAC) instruction to improve computational efficiency.
- The architecture design and integration strategy have been planned to enhance performance of autonomous swarm drone collision avoidance.
- Future work involves completing RTL implementation, synthesis, and simulation to validate functionality and evaluate area, speed, and power improvements.
- We successfully implemented the RV32IM processor with MAC unit integration in Xilinx Vivado tool and got required speed, power, and power.

References

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